

PS / AXI-Lite Control Plane

register banks, status, software ABI

AFE + DAC Control
SPI / bring-up / config_ready

Timing Endpoint Control
clock-source selection / status

Trigger Control
thresholds / delay / modes

analog + frontend

timing and endpoint

AFE Block

5 AFEs $\times$ 8 channels = 40 channels

Endpoint / Timing Subsystem

timestamp, sync, recovered clock, ready

digitized
samples

timestamp, sync, ready
self-trigger and descriptors

Frontend Boundary

LVDS capture, IDELAY, bitslip, alignment valid

aligned channel words

Trigger Fabric

baseline removal, XC filter, trigger gates

triggered sample stream

readout and transport

Two-Lane Multiplexer

shared drain / arbitration

payload stream

Hermes Transport

IPBus / UDP / 10G

debug tap

trigger metadata

records

Peak Descriptor Path

$A_{\max}$, Q , $t_{\max}$, W , N_{obs}

STC3 Record Builder

framed record per channel

descriptor fields

debug / spy path

Spy Trigger + Spybuffer

debug capture, observation memory